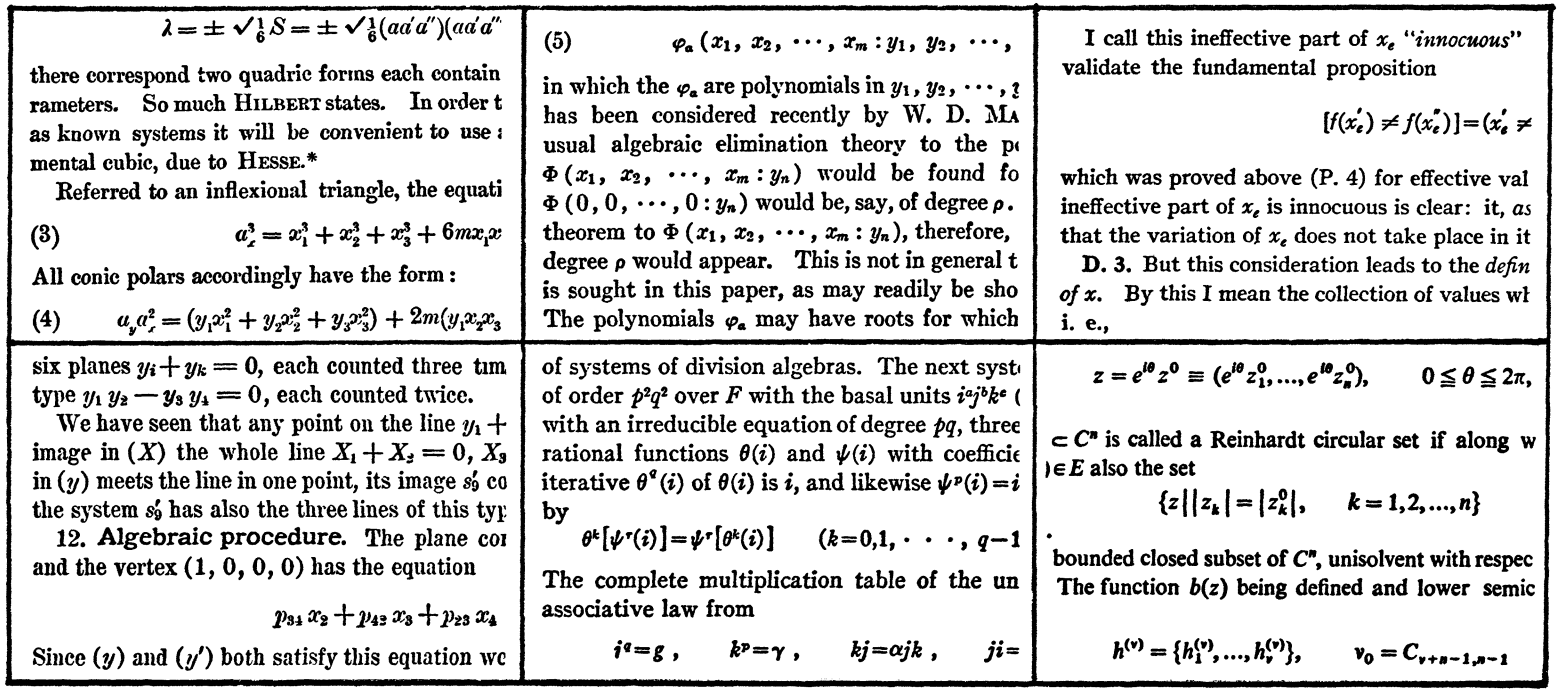


$$\lambda = \pm \sqrt{\tfrac{1}{6}} S = \pm \sqrt{\tfrac{1}{6}} (aa'a'')(aa'a''$$

there correspond two quadric forms each contain
rameters. So much HILBERT states. In order t
as known systems it will be convenient to use
mental cubic, due to HESSE.*

Referred to an inflexional triangle, the equati

(3) $$a_x^3 = x_1^3 + x_2^3 + x_3^3 + 6mx_1x$$

All conic polars accordingly have the form:

(4) $$a_y a_x^2 = (y_1x_1^2 + y_2x_2^2 + y_3x_3^2) + 2m(y_1x_2x_3$$

six planes $y_i + y_k = 0$, each counted three tim
type $y_1 y_2 - y_3 y_4 = 0$, each counted twice.

We have seen that any point on the line $y_1 +$
image in (X) the whole line $X_1 + X_2 = 0$, X_3
in (y) meets the line in one point, its image s_9' co
the system s_9' has also the three lines of this typ

12. **Algebraic procedure.** The plane co
and the vertex (1, 0, 0, 0) has the equation

$$p_{34}x_2 + p_{42}x_3 + p_{23}x_4$$

Since (y) and (y') both satisfy this equation we

(5) $$\varphi_a(x_1, x_2, \cdots, x_m : y_1, y_2, \cdots,$$

in which the φ_a are polynomials in $y_1, y_2, \cdots,$?
has been considered recently by W. D. MA
usual algebraic elimination theory to the p
$\Phi(x_1, x_2, \cdots, x_m : y_n)$ would be found fo
$\Phi(0, 0, \cdots, 0 : y_n)$ would be, say, of degree ρ.
theorem to $\Phi(x_1, x_2, \cdots, x_m : y_n)$, therefore,
degree ρ would appear. This is not in general t
is sought in this paper, as may readily be sho
The polynomials φ_a may have roots for which

of systems of division algebras. The next syst
of order p^2q^2 over F with the basal units $i^a j^b k^c$ (
with an irreducible equation of degree pq, three
rational functions $\theta(i)$ and $\psi(i)$ with coefficie
iterative $\theta^q(i)$ of $\theta(i)$ is i, and likewise $\psi^p(i) = i$
by

$$\theta^k[\psi^r(i)] = \psi^r[\theta^k(i)] \qquad (k = 0,1, \cdots, q-1$$

The complete multiplication table of the un
associative law from

$$i^q = g, \qquad k^p = \gamma, \qquad kj = \alpha jk, \qquad ji =$$

I call this ineffective part of x_e "*innocuous*"
validate the fundamental proposition

$$[f(x_e') \neq f(x_e'')] = (x_e' \neq$$

which was proved above (P. 4) for effective val
ineffective part of x_e is innocuous is clear: it, *as*
that the variation of x_e does not take place in it

D. 3. But this consideration leads to the *defin*
of x. By this I mean the collection of values wh
i. e.,

$$z = e^{i\theta} z^0 \equiv (e^{i\theta} z_1^0, \ldots, e^{i\theta} z_n^0), \qquad 0 \leqq \theta \leqq 2\pi,$$

$\subset C^n$ is called a Reinhardt circular set if along w
$) \in E$ also the set

$$\{z \,|\, |z_k| = |z_k^0|, \qquad k = 1,2,\ldots,n\}$$

bounded closed subset of C^n, unisolvent with respec
The function $b(z)$ being defined and lower semic

$$h^{(\nu)} = \{h_1^{(\nu)}, \ldots, h_\nu^{(\nu)}\}, \qquad \nu_0 = C_{\nu+n-1,n-1}$$

FIGURE 1. A sequence of typographical styles in the AMS *Transactions*:
(a) vol **1** (1900), p. 2; (b) vol **13** (1912), p. 135; (c) vol **23** (1922), p. 216;
(d) vol **25** (1923), p. 10; (e) vol **28** (1926), p. 207; (f) vol **105** (1962), p. 340;

$$0 = r_k x(\sum r_i\alpha_i) - (\sum r_i\alpha_i)xr_k = \sum_{i=1}^{k-1}$$

This element is of lower length. It follows there $i = 1, \cdots, k$. Hence, (a) yields that $r_i = \lambda_i r_k$, λ_i Now $r_k \neq 0$, by the minimality of k, and $\sum\lambda_i$ which we deduce that $\sum\lambda_i\alpha_i = 0$. But the α_i a which is impossible since in particular $\lambda_k = 1$.

THEOREM 7. *Let R be a dense ring of linear t. F be a maximal commutative subfield D. If R_F tion of finite rank over F, then R contains als*

The set N_1 is nowhere dense in Z_1 and thus $N = \rho($

For each $\zeta \in Y - N$ we must prove that f_ζ satisfi be the unique projection in $\{P_d \mid d \in D\}$ such that the algebra $(E\mathscr{A}E)\cdot P_0$ is finite and homogeneou onal abelian projections $E_1, E_2, \ldots, E_n$ such that $(1 \leqq j, k \leqq n)$ be partial isometric operators in $(E\mathscr{A}$

(1) $U_{jk}U_{lm} = \delta_{mj}U_{lk}$, where δ is the Kronecker d
(2) $U_{jk}^* = U_{kj}$; and
(3) $U_{jj} = E_j$,

for all $1 \leqq j, k, l, m \leqq n$. For each A in $(E\mathscr{A}E)\cdot P_0$, t in $\mathscr{Z}_1 P_0$ such that

The algebra P is nearly simple if and only if th

(a) *N is spanned by $a, \cdots, a^{n-k-1}, b_1, \cdots$ $i, j = 1, \cdots, k$.*

(b) *Either $n - k =$ char F with k even or n*

Proof. By Theorem 5.5, there are elements $a, \cdots, a^{n-k-1}, b_1, \cdots, b_k$. Furthermore, $ab_i =$ for all i, j where each α_i, λ_{ij} is in F. From t space of the space spanned by $a^{n-k-1}, b_1, \cdots,$

Assume P is nearly simple. Then there is show that each b_i is in M. To do this, it is nec

:tions in GL(W) and $h_{\alpha\beta}$, $\alpha, \beta \in I$ as coordinate ined by the respective bases chosen above. If α, nction of $\wedge^p$ is the minor of $|g_{ij}|$ determined by columns $\beta(1), \ldots, \beta(p)$. The coordinate ring of $_{\alpha\beta}$ together with $1/\det|h_{\alpha\beta}|$, while that of GL(W) ogether with $1/\det|g_{ij}|$. The coordinate functions , so to show $\wedge^p$ is a morphism it suffices to show nial in g_{ij} and $1/\det|g_{ij}|$. For this, the following

haracter of GL(W) *is an integral power of the*

of Q, i.e.

$i) = 0$ for every $x \in A$ for which $x(Q) = 0\}$.

'or m_A is equivalent to the one induced by the :

$\{|x(z)|: x \in A, \|x\| \leqslant 1$ and $x(w) = 0\}$.

:present the open unit disk in the complex plane, $\mathbf{C}$, t polydisk in n-dimensional complex space $\mathbf{C}^n$. T^n oundary of D^n, i.e.

:onverges pathwise to X^λ, and uniformly for $t \in$ for which X_i is the (last) minimum of Y^λ, let Y_i^λ, ılues of Y^λ, and T_i^λ the interjump times for Y^λ st an i such that $Y_i^\lambda = T_i^\lambda = \infty$. Notice that Y_Q^λ i ıd that as $\varepsilon \to 0$, Y_Q^λ converges to $I^\lambda = \inf_s X_s^\lambda$. Le ts of $(-\infty, \infty)$. Then, for example, if $i \geqslant 1$

$-i \in B, Y_{Q+k}^\lambda - Y_Q^\lambda \in C, T_{Q+k}^\lambda \in D, N > Q > i\}$

$\in A, T_{l-i}^\lambda \in B, Y_{l+k}^\lambda - Y_l^\lambda \in C, T_{l+k}^\lambda \in D, N > Q$

FIGURE 1 [continued]:

(g) vol **114** (1965), p. 216; (h) vol **125** (1966), p. 38; (i) vol **169** (1972), p. 232;
(j) vol **179** (1973), p. 314; (k) vol **199** (1974), p. 370; (l) vol **225** (1977), p. 372.